

Week5-EP

September 6, 2026

1 Online Job Market Analytics

1.1 Web Scraping using Selenium

1.1.1 Problem Statement

A recruitment agency wants to analyze job opportunities posted on a dynamic job portal.

The program will:

- Scrape job listings.
- Extract Job Title, Company Name, Location and Salary.
- Store the collected data in a Pandas DataFrame.
- Save the dataset as a CSV file.
- Analyze job demand by location.
- Visualize location-wise job demand.

Website Used: <https://www.naukri.com>

```
[1]: pip install -q selenium pandas matplotlib
```

Note: you may need to restart the kernel to use updated packages.

```
[2]: import time
import pandas as pd
import matplotlib.pyplot as plt

from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from selenium.common.exceptions import NoSuchElementException, TimeoutException
```

```
[3]: JOB_ROLE = "data scientist"
LOCATION = "bangalore"
MAX_JOBS = 30

print("Job Role:", JOB_ROLE)
print("Location:", LOCATION)
print("Maximum Jobs:", MAX_JOBS)
```

Job Role: data scientist
Location: bangalore
Maximum Jobs: 30

```
[4]: options = Options()

driver = webdriver.Firefox(options=options)

driver.set_window_size(1400, 900)

print("Firefox started successfully.")
```

Firefox started successfully.

```
[5]: def make_slug(text):
      return "-".join(text.lower().strip().split())

job_slug = make_slug(JOB_ROLE)
location_slug = make_slug(LOCATION)

url = f"https://www.naukri.com/{job_slug}-jobs-in-{location_slug}"

print("Search URL:", url)
```

Search URL: https://www.naukri.com/data-scientist-jobs-in-bangalore

```
[6]: driver.get(url)

time.sleep(5)

print("Page Title:", driver.title)
print("Current URL:", driver.current_url)
```

Page Title: Data Scientist Jobs In Bangalore - 7018 Data Scientist Job Vacancies In Bangalore - Naukri.com

Current URL: https://www.naukri.com/data-scientist-jobs-in-bangalore

```
[7]: print("Website Title:")
      print(driver.title)

      print("\nURL:")
      print(driver.current_url)
```

Website Title:

Data Scientist Jobs In Bangalore - 7018 Data Scientist Job Vacancies In Bangalore - Naukri.com

URL:

https://www.naukri.com/data-scientist-jobs-in-bangalore

```
[8]: for i in range(3):
        driver.execute_script(
            "window.scrollTo(0, document.body.scrollHeight);"
        )
        time.sleep(2)

    driver.execute_script(
        "window.scrollTo(0, 0);"
    )

    print("Page scrolling completed.")
```

Page scrolling completed.

```
[9]: cards = driver.find_elements(
        By.CSS_SELECTOR,
        "div.cust-job-tuple"
    )

    print("Number of job cards found:", len(cards))
```

Number of job cards found: 20

```
[10]: print(cards[0].text)
```

Data Scientist
 Aditya Birla Management Corporation (ABMCPL)
 4.1
 2366 Reviews
 7-10 Yrs
 Bengaluru
 Position Title - Data Scientist Job Location - BengaluruJob Purpose The role focuses o...
 Machine LearningDeep LearningPythonPredictive ModelingPysparkArtificial IntelligenceNatural Language ProcessingNeural Networks
 1 week ago
 Save

```
[11]: def get_text(card, selectors, default="Not Available"):

        for selector in selectors:

            try:
                element = card.find_element(
                    By.CSS_SELECTOR,
                    selector
                )

                text = element.text.strip()
```

```

        if text:
            return text

    except NoSuchElementException:
        pass

    return default

```

```
[12]: jobs = []
```

```
[13]: for card in cards[:MAX_JOBS]:

    title = get_text(
        card,
        [
            "a.title",
            "[class*='title']"
        ]
    )

    company = get_text(
        card,
        [
            "a.comp-name",
            ".comp-name"
        ]
    )

    location = get_text(
        card,
        [
            "span.locWdth",
            ".locWdth",
            "[class*='location']"
        ]
    )

    salary = get_text(
        card,
        [
            "span.sal",
            ".sal",
            "[class*='salary']"
        ],
        default="Not Disclosed"
    )

```

```

jobs.append({
    "Job Title": title,
    "Company Name": company,
    "Location": location,
    "Salary": salary
})

print("Jobs scraped:", len(jobs))

```

Jobs scraped: 20

```
[14]: jobs[:3]
```

```

[14]: [{'Job Title': 'Data Scientist',
      'Company Name': 'Aditya Birla Management Corporation (ABMCPL)',
      'Location': 'Bengaluru',
      'Salary': 'Not Disclosed'},
      {'Job Title': 'Data Scientist/Software Developer',
      'Company Name': 'Xylem Water Solutions',
      'Location': 'Bengaluru',
      'Salary': 'Not Disclosed'},
      {'Job Title': 'Data Scientist III',
      'Company Name': 'Walmart',
      'Location': 'Bengaluru',
      'Salary': 'Not Disclosed'}]

```

```

[15]: df = pd.DataFrame(jobs)

df

```

```

[15]:
      Job Title \
0      Data Scientist
1  Data Scientist/Software Developer
2      Data Scientist III
3      Data Scientist
4  Data Scientist | Python | GenAI | LLM | RAG
5      Machine Learning Data Scientist
6      Advanced Data Scientist
7      Data Scientist
8  Data Scientist | GenAI | Agentic AI | LLM | RAG
9      Data Scientist AIML
10 Data Scientist/Python/MLOps- 4th Sep(Fri)-6+ y...
11      Data scientist III
12      Data Scientist
13      Data Scientist/ Data Analyst
14      Walk-in || Data Scientist
15      Data Scientist (Fulltime or Contract)

```

16 Advanced Data Scientist
 17 Python with data scientist + Gen AI/ LangChain...
 18 Data Scientist- MLOPS
 19 Data Scientist/AI-ML Engg

	Company Name \
0	Aditya Birla Management Corporation (ABMCPL)
1	Xylem Water Solutions
2	Walmart
3	Genpact
4	Infosys
5	Hsbc
6	ExxonMobil
7	VOIS
8	Infosys
9	Tata Consultancy Services
10	Tata Consultancy Services
11	Swiggy
12	Maruti Suzuki
13	PwC
14	Capgemini
15	Carnation Infotech
16	Antuit
17	Cloudxtreme
18	Persistent
19	Tavant

	Location	Salary
0	Bengaluru	Not Disclosed
1	Bengaluru	Not Disclosed
2	Bengaluru	Not Disclosed
3	Bengaluru	20-30 Lacs PA
4	Hybrid - Hyderabad, Chennai, Bengaluru	Not Disclosed
5	Bengaluru	Not Disclosed
6	Bengaluru	Not Disclosed
7	Pune, Bengaluru	Not Disclosed
8	Hybrid - Pune, Bengaluru, Delhi / NCR	Not Disclosed
9	Hyderabad, Chennai, Bengaluru	Not Disclosed
10	Pune, Bengaluru, Delhi / NCR	Not Disclosed
11	Bengaluru	Not Disclosed
12	Bengaluru	Not Disclosed
13	Hybrid - Hyderabad, Gurugram, Bengaluru	Not Disclosed
14	Hybrid - Bengaluru	Not Disclosed
15	Remote	15-20 Lacs PA
16	Bengaluru	Not Disclosed
17	Hyderabad, Chennai, Bengaluru	14-20 Lacs PA
18	Bengaluru	0-18 Lacs PA

```
[16]: df = df.drop_duplicates().reset_index(drop=True)

print("Total unique jobs:", len(df))

df
```

Total unique jobs: 20

```
[16]:
```

	Job Title \
0	Data Scientist
1	Data Scientist/Software Developer
2	Data Scientist III
3	Data Scientist
4	Data Scientist Python GenAI LLM RAG
5	Machine Learning Data Scientist
6	Advanced Data Scientist
7	Data Scientist
8	Data Scientist GenAI Agentic AI LLM RAG
9	Data Scientist AIML
10	Data Scientist/Python/MLOps- 4th Sep(Fri)-6+ y...
11	Data scientist III
12	Data Scientist
13	Data Scientist/ Data Analyst
14	Walk-in Data Scientist
15	Data Scientist (Fulltime or Contract)
16	Advanced Data Scientist
17	Python with data scientist + Gen AI/ LangChain...
18	Data Scientist- MLOPS
19	Data Scientist/AI-ML Engg

	Company Name \
0	Aditya Birla Management Corporation (ABMCPL)
1	Xylem Water Solutions
2	Walmart
3	Genpact
4	Infosys
5	Hsbc
6	ExxonMobil
7	VOIS
8	Infosys
9	Tata Consultancy Services
10	Tata Consultancy Services
11	Swiggy
12	Maruti Suzuki
13	PwC
14	Capgemini

```

15          Carnation Infotech
16          Antuit
17          Cloudxtreme
18          Persistent
19          Tavant

```

		Location	Salary
0		Bengaluru	Not Disclosed
1		Bengaluru	Not Disclosed
2		Bengaluru	Not Disclosed
3		Bengaluru	20-30 Lacs PA
4	Hybrid - Hyderabad, Chennai,	Bengaluru	Not Disclosed
5		Bengaluru	Not Disclosed
6		Bengaluru	Not Disclosed
7		Pune, Bengaluru	Not Disclosed
8	Hybrid - Pune, Bengaluru, Delhi / NCR		Not Disclosed
9	Hyderabad, Chennai,	Bengaluru	Not Disclosed
10	Pune, Bengaluru, Delhi / NCR		Not Disclosed
11		Bengaluru	Not Disclosed
12		Bengaluru	Not Disclosed
13	Hybrid - Hyderabad, Gurugram,	Bengaluru	Not Disclosed
14	Hybrid -	Bengaluru	Not Disclosed
15		Remote	15-20 Lacs PA
16		Bengaluru	Not Disclosed
17	Hyderabad, Chennai,	Bengaluru	14-20 Lacs PA
18		Bengaluru	0-18 Lacs PA
19		Bengaluru	Not Disclosed

```
[17]: df.head(10)
```

```

[17]:
0          Data Scientist
1  Data Scientist/Software Developer
2          Data Scientist III
3          Data Scientist
4  Data Scientist | Python | GenAI | LLM | RAG
5          Machine Learning Data Scientist
6          Advanced Data Scientist
7          Data Scientist
8  Data Scientist | GenAI | Agentic AI | LLM | RAG
9          Data Scientist AIML

          Company Name \
0  Aditya Birla Management Corporation (ABMCPL)
1          Xylem Water Solutions
2          Walmart
3          Genpact

```

4	Infosys
5	Hsbc
6	ExxonMobil
7	VOIS
8	Infosys
9	Tata Consultancy Services

	Location	Salary
0	Bengaluru	Not Disclosed
1	Bengaluru	Not Disclosed
2	Bengaluru	Not Disclosed
3	Bengaluru	20-30 Lacs PA
4	Hybrid - Hyderabad, Chennai, Bengaluru	Not Disclosed
5	Bengaluru	Not Disclosed
6	Bengaluru	Not Disclosed
7	Pune, Bengaluru	Not Disclosed
8	Hybrid - Pune, Bengaluru, Delhi / NCR	Not Disclosed
9	Hyderabad, Chennai, Bengaluru	Not Disclosed

```
[18]: print("Dataset Shape:", df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nTotal Records:", len(df))
```

Dataset Shape: (20, 4)

Column Names:
['Job Title', 'Company Name', 'Location', 'Salary']

Total Records: 20

```
[19]: print("Salary Information:")

print(df["Salary"].value_counts())
```

Salary Information:

Salary	
Not Disclosed	16
20-30 Lacs PA	1
15-20 Lacs PA	1
14-20 Lacs PA	1
0-18 Lacs PA	1

Name: count, dtype: int64

```
[20]: df.to_csv(
    "job_market_data.csv",
```

```

        index=False
    )

    print("CSV file saved successfully.")

```

CSV file saved successfully.

```

[21]: csv_data = pd.read_csv(
        "job_market_data.csv"
    )

    csv_data.head()

```

```

[21]:

```

	Job Title \
0	Data Scientist
1	Data Scientist/Software Developer
2	Data Scientist III
3	Data Scientist
4	Data Scientist Python GenAI LLM RAG

	Company Name \
0	Aditya Birla Management Corporation (ABMCPL)
1	Xylem Water Solutions
2	Walmart
3	Genpact
4	Infosys

	Location	Salary
0	Bengaluru	Not Disclosed
1	Bengaluru	Not Disclosed
2	Bengaluru	Not Disclosed
3	Bengaluru	20-30 Lacs PA
4	Hybrid - Hyderabad, Chennai, Bengaluru	Not Disclosed

```

[22]: cities = [
        "bangalore",
        "hyderabad",
        "pune",
        "chennai",
        "mumbai",
        "delhi"
    ]

    print("Cities to analyze:")
    print(cities)

```

Cities to analyze:

```
['bangalore', 'hyderabad', 'pune', 'chennai', 'mumbai', 'delhi']
```

```
[23]: all_jobs=[]
```

```
[24]: for city in cities:

    city_slug = make_slug(city)

    url = f"https://www.naukri.com/{job_slug}-jobs-in-{city_slug}"

    print("\nScraping:", city)

    driver.get(url)
    time.sleep(4)

    city_cards = driver.find_elements(
        By.CSS_SELECTOR,
        "div.cust-job-tuple"
    )

    print("Jobs found:", len(city_cards))

    for card in city_cards[:10]:

        title = get_text(
            card,
            [
                "a.title",
                "[class*='title']"
            ]
        )

        company = get_text(
            card,
            [
                "a.comp-name",
                ".comp-name"
            ]
        )

        salary = get_text(
            card,
            [
                "span.sal",
                ".sal",
                "[class*='salary']"
            ],
            default="Not Disclosed"
        )
```

```

all_jobs.append({
    "Job Title": title,
    "Company Name": company,
    "Location": city.title(),
    "Salary": salary
})

```

Scraping: bangalore
Jobs found: 20

Scraping: hyderabad
Jobs found: 20

Scraping: pune
Jobs found: 20

Scraping: chennai
Jobs found: 20

Scraping: mumbai
Jobs found: 21

Scraping: delhi
Jobs found: 21

```

[25]: city_df = pd.DataFrame(all_jobs)

city_df = city_df.drop_duplicates().reset_index(drop=True)

print("Total jobs collected:", len(city_df))

city_df.head(10)

```

Total jobs collected: 60

```

[25]:
      Job Title \
0          Data Scientist
1  Data Scientist/Software Developer
2          Data Scientist III
3          Data Scientist
4  Data Scientist | Python | GenAI | LLM | RAG
5          Machine Learning Data Scientist
6          Advanced Data Scientist
7          Data Scientist
8  Data Scientist | GenAI | Agentic AI | LLM | RAG
9          Data Scientist AIML

```

	Company Name	Location	Salary
0	Aditya Birla Management Corporation (ABMCPL)	Bangalore	Not Disclosed
1	Xylem Water Solutions	Bangalore	Not Disclosed
2	Walmart	Bangalore	Not Disclosed
3	Genpact	Bangalore	20-30 Lacs PA
4	Infosys	Bangalore	Not Disclosed
5	Hsbc	Bangalore	Not Disclosed
6	ExxonMobil	Bangalore	Not Disclosed
7	VOIS	Bangalore	Not Disclosed
8	Infosys	Bangalore	Not Disclosed
9	Tata Consultancy Services	Bangalore	Not Disclosed

```
[26]: location_analysis = (
        city_df["Location"]
        .value_counts()
        .reset_index()
    )

    location_analysis.columns = [
        "Location",
        "Number of Jobs"
    ]

    location_analysis
```

```
[26]:      Location  Number of Jobs
0  Bangalore         10
1  Hyderabad         10
2    Pune         10
3  Chennai         10
4  Mumbai         10
5   Delhi         10
```

```
[27]: plt.figure(figsize=(10, 5))

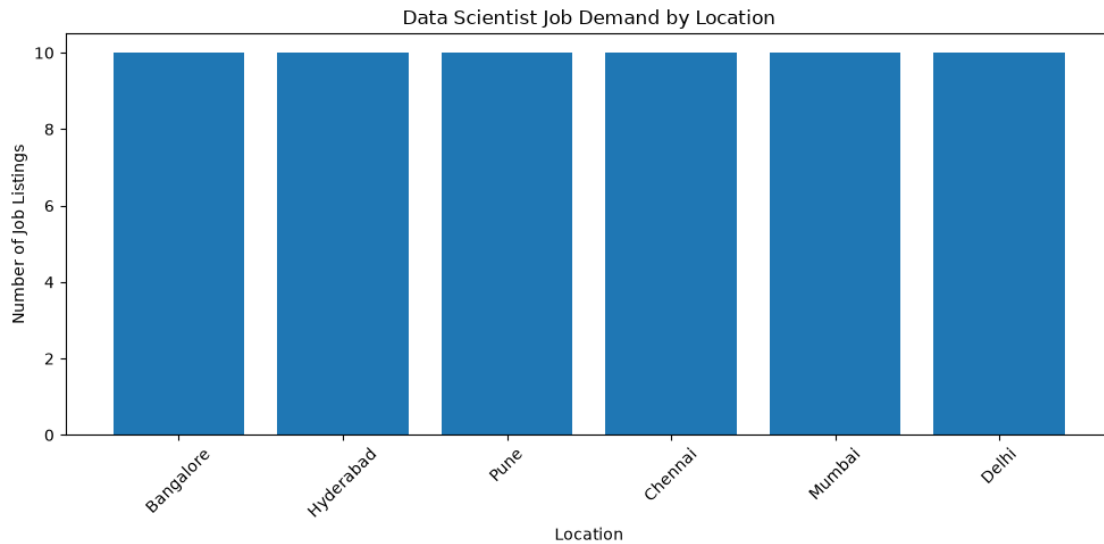
plt.bar(
    location_analysis["Location"],
    location_analysis["Number of Jobs"]
)

plt.title("Data Scientist Job Demand by Location")
plt.xlabel("Location")
plt.ylabel("Number of Job Listings")

plt.xticks(rotation=45)
```

```
plt.tight_layout()
```

```
plt.show()
```



```
[28]: plt.figure(figsize=(10, 5))

plt.bar(
    location_analysis["Location"],
    location_analysis["Number of Jobs"]
)

plt.title("Data Scientist Job Demand by Location")
plt.xlabel("Location")
plt.ylabel("Number of Job Listings")

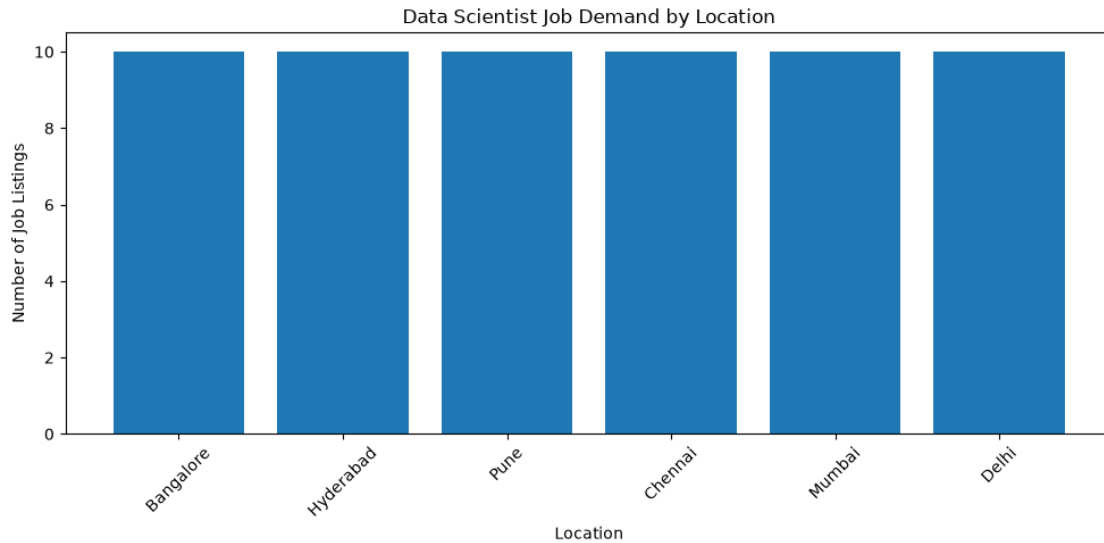
plt.xticks(rotation=45)

plt.tight_layout()

plt.savefig(
    "job_demand_by_location.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()

print("Graph saved successfully.")
```



Graph saved successfully.

```
[29]: city_df.to_csv(
        "job_market_multi_city.csv",
        index=False
    )

    print("Multi-city dataset saved successfully.")
```

Multi-city dataset saved successfully.

```
[30]: print("ONLINE JOB MARKET ANALYTICS")
    print("-" * 35)

    print("Job Role:", JOB_ROLE)
    print("Total Jobs Analyzed:", len(city_df))
    print("Cities Analyzed:", len(location_analysis))

    print("\nLocation-wise Job Demand:")
    print(location_analysis.to_string(index=False))
```

ONLINE JOB MARKET ANALYTICS

Job Role: data scientist
 Total Jobs Analyzed: 60
 Cities Analyzed: 6

Location-wise Job Demand:

Location	Number of Jobs
Bangalore	10

Hyderabad	10
Pune	10
Chennai	10
Mumbai	10
Delhi	10

```
[31]: driver.quit()

print("Browser closed successfully.")
```

Browser closed successfully.

```
[ ]:
```